Driver
Detailed System Role & Workflow

Drivers execute trips, operate buses, and interact with the AI safety and operational systems.

1. Primary Responsibilities

- Execute assigned trips
- Follow operational schedules
- Interact with onboard safety systems
- Receive alerts and warnings
- Maintain operational compliance
- Report incidents or emergencies

2. Full Operational Flow

Drivers interact directly with onboard smart systems.

When a driver starts a trip:
- MDVR activates
- GPS tracking begins
- DMS camera monitors driver behavior
- ADAS monitors road conditions
- Passenger systems activate

The driver receives:
- Voice alerts
- Collision warnings
- Fatigue warnings
- Lane departure alerts

All violations and safety events are logged into the ERP.

3. System Interactions

- Interacts with ERP modules
- Interacts with live operational data
- Uses dashboards and reports

- Triggers workflow actions
- Generates system events
- Consumes real-time information
- Affects operational flow and business logic

4. Real-Time Data Impact

This role directly interacts with the real-time transportation ecosystem.

Actions performed by this role can affect:
- Bus operations
- Driver assignments
- Camera monitoring
- GPS tracking
- Financial workflows
- Maintenance workflows
- AI safety alerts
- Passenger information systems

Because the platform is interconnected, actions performed by one role automatically affect other departments and operational flows.

5. Important Developer Considerations

- Role-based access control (RBAC) is critical
- Permissions must be modular and scalable
- Realtime updates are required
- Audit logs should track all actions
- Cross-department workflow integration is necessary
- Dashboard views should differ by role
- Notification systems should be role-aware
- Operational events should be linked across modules

6. Final Assessment

The Driver role is a critical part of the SEUM intelligent transportation ecosystem.

This role interacts not only with traditional ERP workflows, but also with:
- Real-time transportation operations
- AI safety systems

- GPS tracking infrastructure
- Smart bus devices
- Live camera monitoring
- Financial and operational analytics

The system must therefore be designed with interconnected workflows and scalable permission architecture.